

Lab 2: Hierarchical Component Design

Requirements Not Met

N/A

Problems Encountered

Figuring out how to use components I made to make other circuits

Applications

Switch between multiple outputs based on simple inputs

Leave a page break between the previous section and the Pre-Lab Questions. Delete this message before submitting.

Pre-Lab Questions

<Answer all Pre-Lab questions here, insert any tables, diagrams, images in this section>

3 B)

B3	B2	B1	B0	OP	G	F	E	D	C	B	A
0	0	0	0	1	1	0	0	0	0	0	0
0	0	0	1	1	1	1	1	1	0	0	1
0	0	1	0	1	0	1	0	0	1	0	0
0	0	1	1	1	0	1	1	0	0	0	0
0	1	0	0	1	0	0	1	1	0	0	1
0	1	0	1	1	0	0	1	0	0	1	0
0	1	1	0	1	0	0	0	0	0	1	0
0	1	1	1	1	1	1	1	1	0	0	0
1	0	0	0	1	0	0	0	0	0	0	0
1	0	0	1	1	0	0	1	0	0	0	1
1	0	1	0	1	0	0	0	1	0	0	0
1	0	1	1	1	0	0	0	0	0	1	1
1	1	0	0	1	1	0	0	0	1	1	0
1	1	0	1	1	0	1	0	0	0	0	1
1	1	1	0	1	0	0	0	0	1	1	0
1	1	1	1	1	0	0	0	1	1	1	0

D k-map:

		B1 B0			
		00	01	11	10
B3 B2	00	0	1	0	0
	01	1	0	1	0
11	11	0	0	1	0
	10	0	0	0	1

$B_2 B_1 B_0 + \overline{B_3} \overline{B_2} \overline{B_1} B_0 + \overline{B_3} B_2 \overline{B_1} \overline{B_0} + B_3 B_2 B_1 \overline{B_0}$
 $+ \overline{B_3} B_2 \overline{B_1} \overline{B_0} + B_3 \overline{B_2} B_1 \overline{B_0}$





